REYHAN DEVARA

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PROFESSIONAL SUMMARY

AI and Embedded Systems Engineer working across real-time computer vision, local AI workflows, and hardware integration. Experienced in developing custom ESP32 and Arduino firmware, PCB schematics, and interactive sensor systems, as well as building vision pipelines using YOLO, OpenCV, and MediaPipe. Skilled in delivering end-to-end prototypes by connecting hardware with web, Unity, and Python applications via REST APIs, WebSockets, and MQTT.

WORK EXPERIENCE

Unictive — West Jakarta, Jakarta, Indonesia — Jan 2025 – Present

Embedded Systems Engineer

Design and develop embedded systems for interactive installations and gamification projects, covering firmware, hardware prototyping, PCB design, and software integration.

- Develop ESP32 and Arduino firmware using PlatformIO and Arduino IDE, integrating sensors and actuators for real-time interaction.
- Design schematics and custom PCBs using EasyEDA, including circuit assembly, soldering, and hardware prototyping.
- Integrate embedded systems with Unity, web, and Python applications via HTTP/REST APIs, sockets, Bluetooth, and wired interfaces.
- Conduct technical research, rapid prototyping, and Proof of Concept (PoC) development to validate project solutions.
- Perform firmware debugging, hardware testing, troubleshooting, and system validation to ensure reliable operation.

Unictive — West Jakarta, Jakarta, Indonesia — Oct 2024 – Present

AI Engineer

Develop and integrate real-time AI and computer vision solutions for interactive installations, gamification, and web applications, including technical research and Proof of Concept (PoC) development.

- Build real-time computer vision pipelines using YOLO (Ultralytics), OpenCV, and MediaPipe for object detection, tracking, and pose/keypoint detection.
- Prepare and manage custom datasets using Roboflow, including annotation, augmentation, and data splitting, along with custom YOLO model training and evaluation.
- Develop local AI inference services using Python, FastAPI, and Flask, integrating AI models with Unity applications and web platforms.
- Research and prototype generative AI workflows using ComfyUI for image generation and AI-powered image processing.
- Explore local LLMs and AI agents using Ollama, Hermes, and LangChain, including MCP, external tool integration, and RAG with Chroma Vector Database.

PT. Syergie Indo Prima — Bandung, West Java, Indonesia — Jan 2022 – Mar 2022

Electrical Engineering Internship

PT. Syergie Indo Prima is an Indonesian consultancy specializing in energy distribution, submarine services, and geotechnical solutions.

- Designed 3+ PCB layouts using Eagle for electronic system development.
- Developed a monitoring and control system GUI using QML and Python, enabling real-time system interaction with data communication via MQTT.

EDUCATION

Universitas Padjadjaran — Sumedang, West Java, Indonesia Aug 2019 – Apr 2024

Bachelor of Electrical Engineering • GPA: 3.72 / 4.00

- Authored and presented a research paper on real-time object edge detection for RPLiDAR sensor performance enhancement, presented at the 4th International Conference on MECnIT 2021.
- Final thesis: "SAVIBot: Virtual Assistant Bot for Intelligent Vehicle Interaction and Control", focusing on the design and development of a chatbot using BERT for natural language understanding.

SKILLS

AI & Computer Vision: YOLO (Ultralytics), OpenCV, MediaPipe, Roboflow, ComfyUI, Inspyrenet, GFPGAN

LLM & Agents: Ollama, Hermes, LangChain, Chroma Vector DB, RAG, Model Context Protocol (MCP), BERT

Programming Languages: Python, Embedded C/C++, JavaScript, SQL

Web & Backend: FastAPI, Flask, REST APIs, WebSockets, PostgreSQL, SQLite

Embedded & Hardware: ESP32, Arduino, PlatformIO, Arduino IDE, FreeRTOS, Sensors & Actuators, Circuit Design, Soldering & Prototyping, PCB Design (EasyEDA, Eagle)

Protocols & Communication: MQTT, Bluetooth, Sockets, HTTP/REST, Serial

Tools & Other: Git, QML, Unity Integration

ACHIEVEMENTS & AWARDS

- **Problem Solver Award** — Unicverse Day (2026)
- **2nd Place in AI Mastery** — Habibie TechFest 2022 (Orbit Future Academy)